

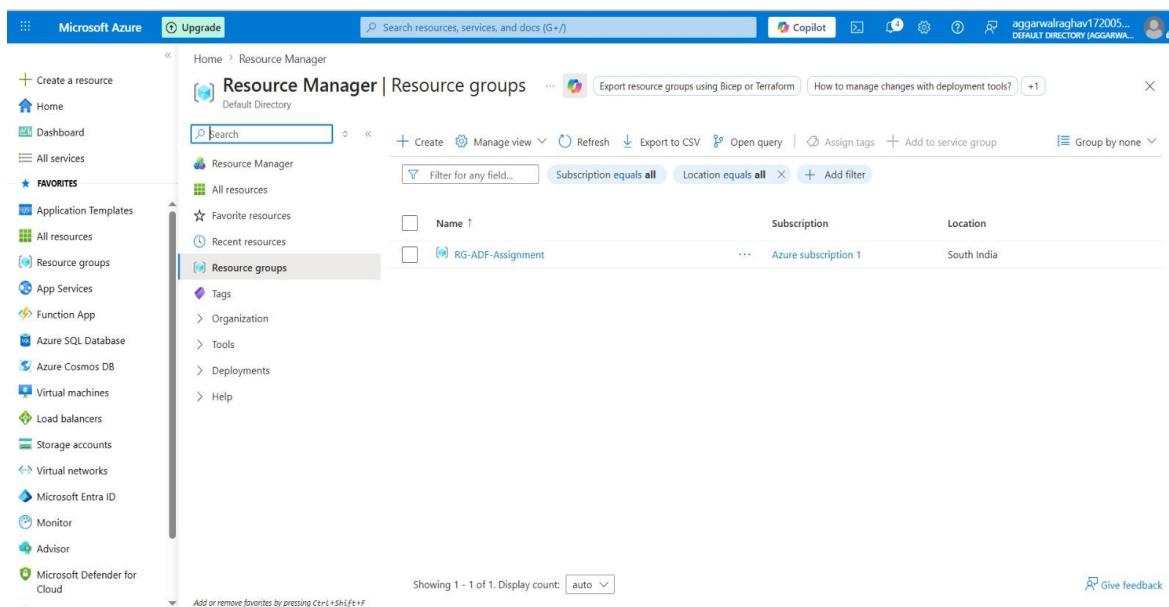
WEEK 4 ASSIGNMENT

INTRODUCTION

This assignment focuses on the implementation of Azure cloud services, including Azure Storage Account, Blob Storage, Azure Data Factory (ADF), and IAM roles. A data pipeline was created to copy a CSV file from a source container to a destination container using the Copy Data activity. The assignment also demonstrates the use of Linked Services, Datasets, and Get Metadata activity for data integration and management in Azure.

TASK 1 – Create Resource Group

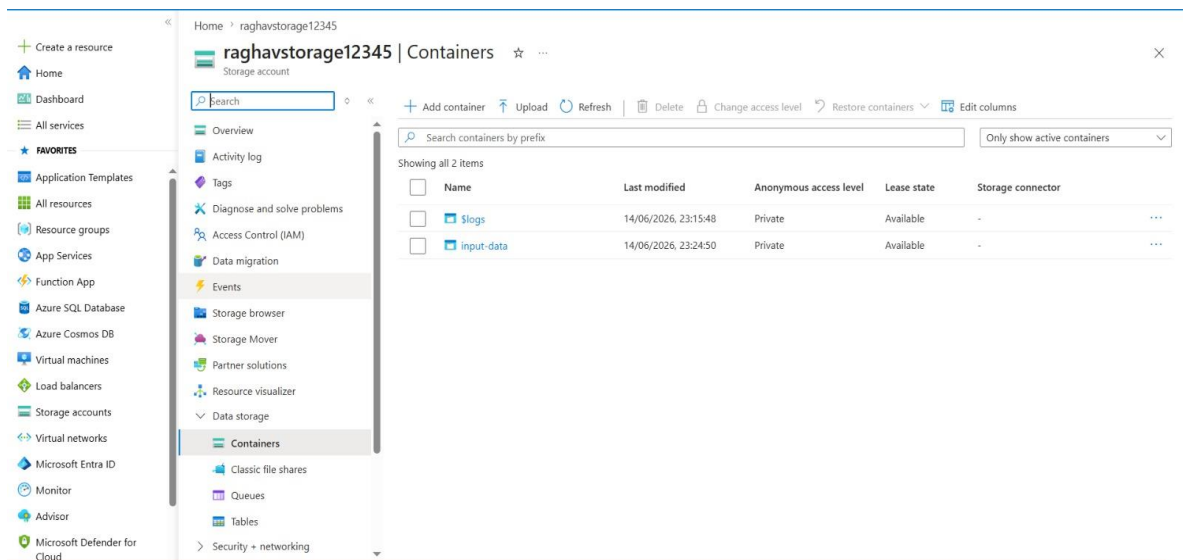
- Resource Group Name: RG-ADF-Assignment
- Region: South India



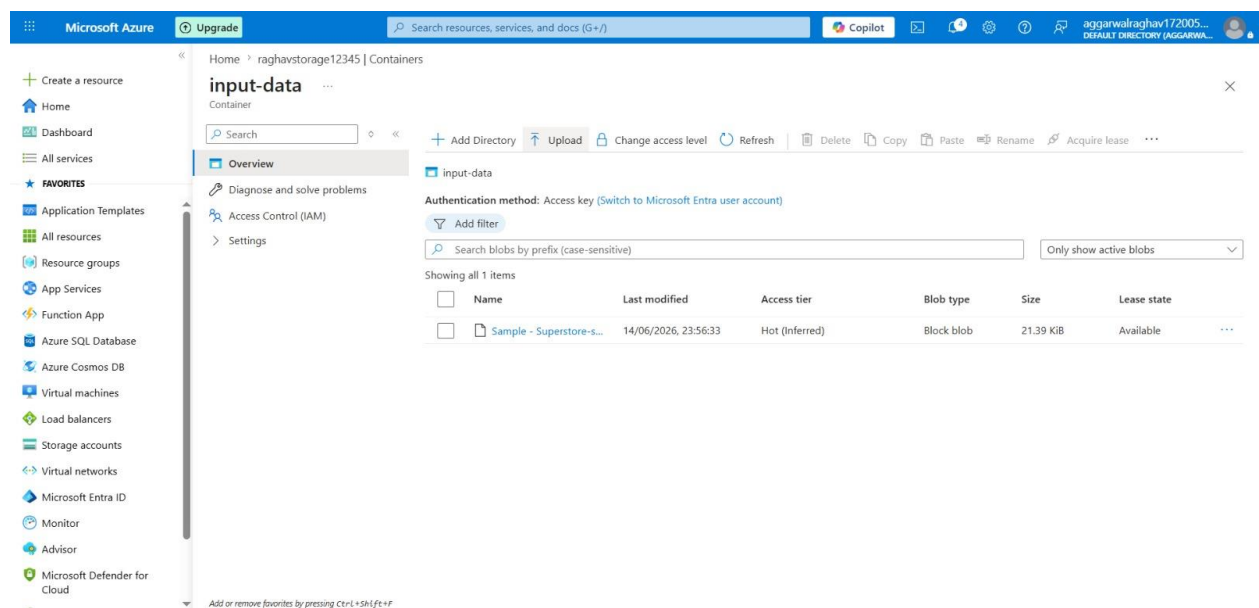
TASK 2 – Create Storage Account And Container

Configuration

- Storage Account Name: raghavstorage12345
- Performance: Standard
- Redundancy: Locally Redundant Storage (LRS)
- Region: South India



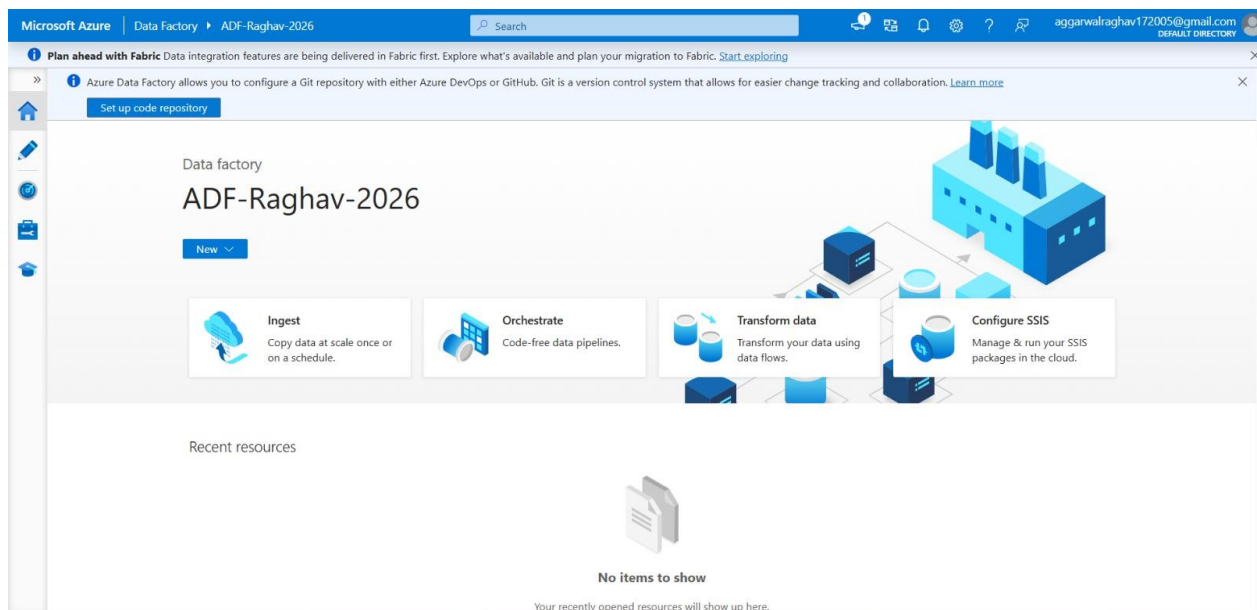
Upload Source CSV File



TASK 3 – ADF Basics

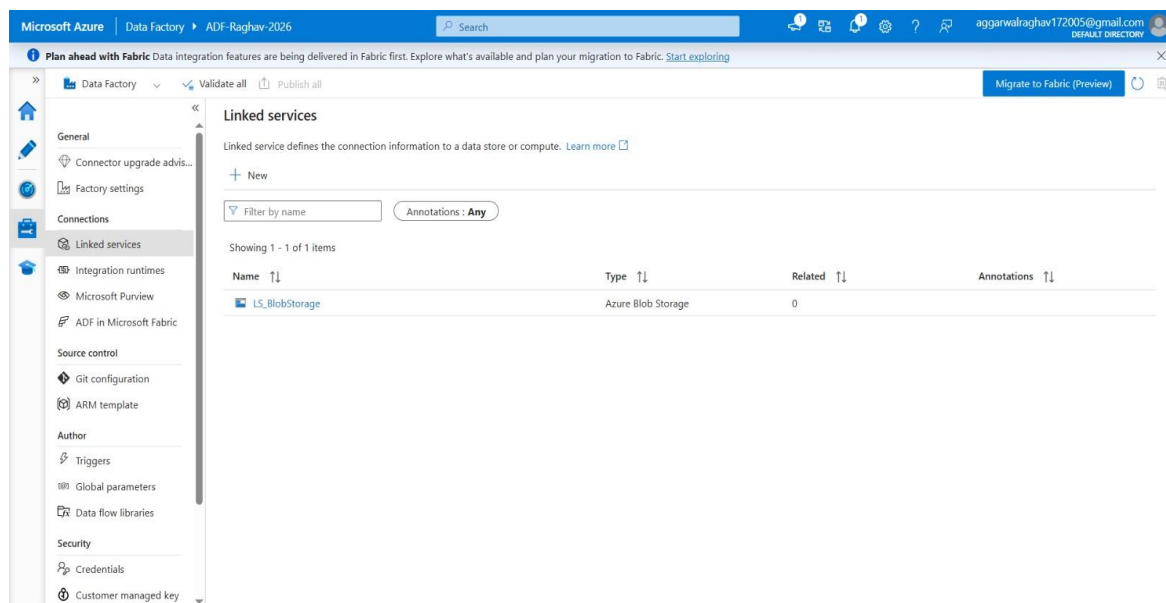
- **Creation of Azure Data Factory**

Azure Data Factory (ADF) is a cloud-based data integration service provided by Microsoft Azure. It is used to create, schedule, and monitor data workflows. In this task, an Azure Data Factory instance was created for data movement and processing activities.



- **Create Linked Service**

A **Linked Service** acts as a connection between Azure Data Factory and external data sources. In this assignment, a Linked Service named **LS_BlobStorage** was created to establish a connection between Azure Data Factory and the Azure Blob Storage account. This connection allows the pipeline to securely access files stored in the storage account.

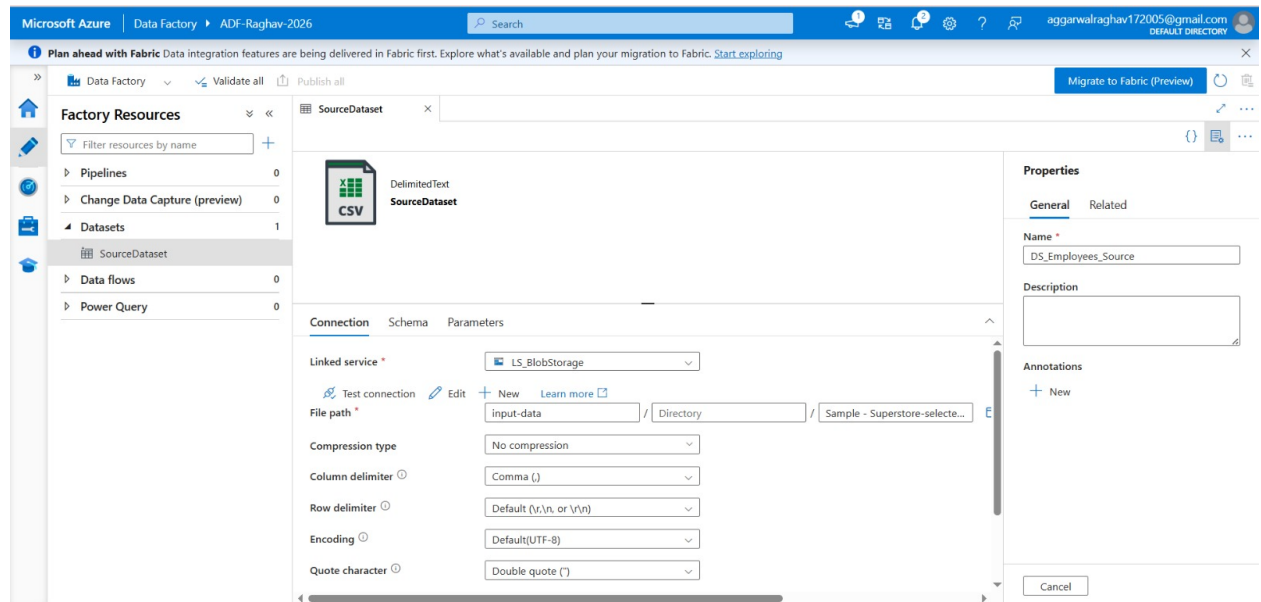


- **Create Source and Destination Datasets**

A Dataset represents the structure and location of data that will be used in a pipeline.

- Name: SourceDataset
- Container: input-data
- File: Sample – Superstore-selected.csv

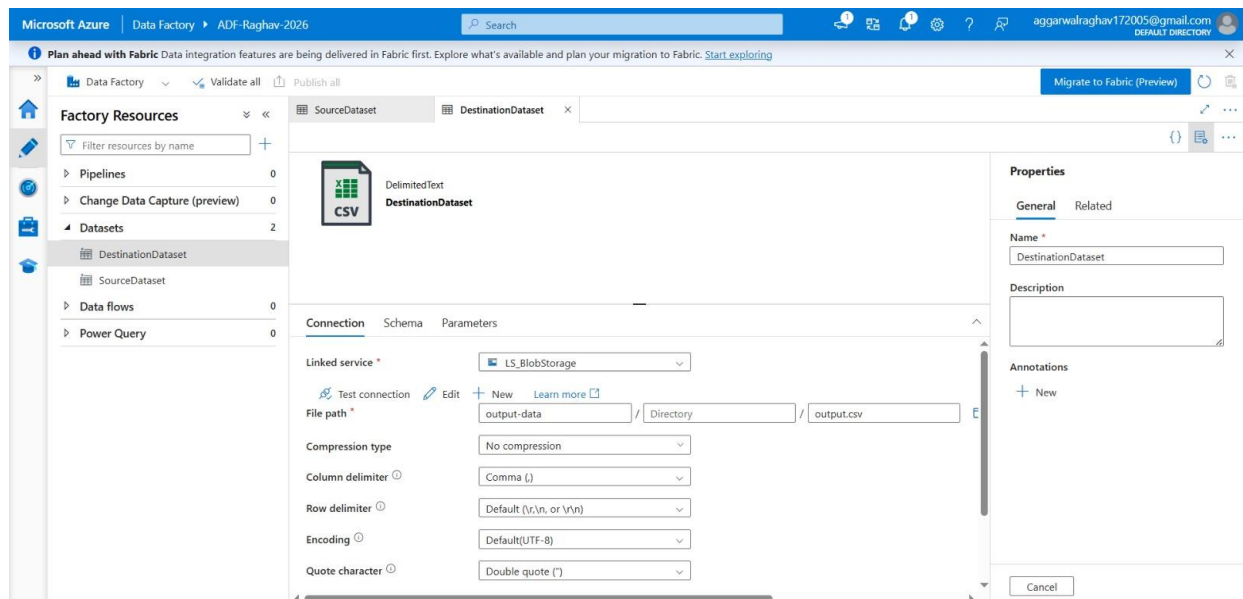
This dataset represents the input file that will be read by the pipeline.



Destination Dataset

- Name: **DestinationDataset**
- Container: **output**
- File: **output.csv**

This dataset represents the destination where the copied data will be stored.



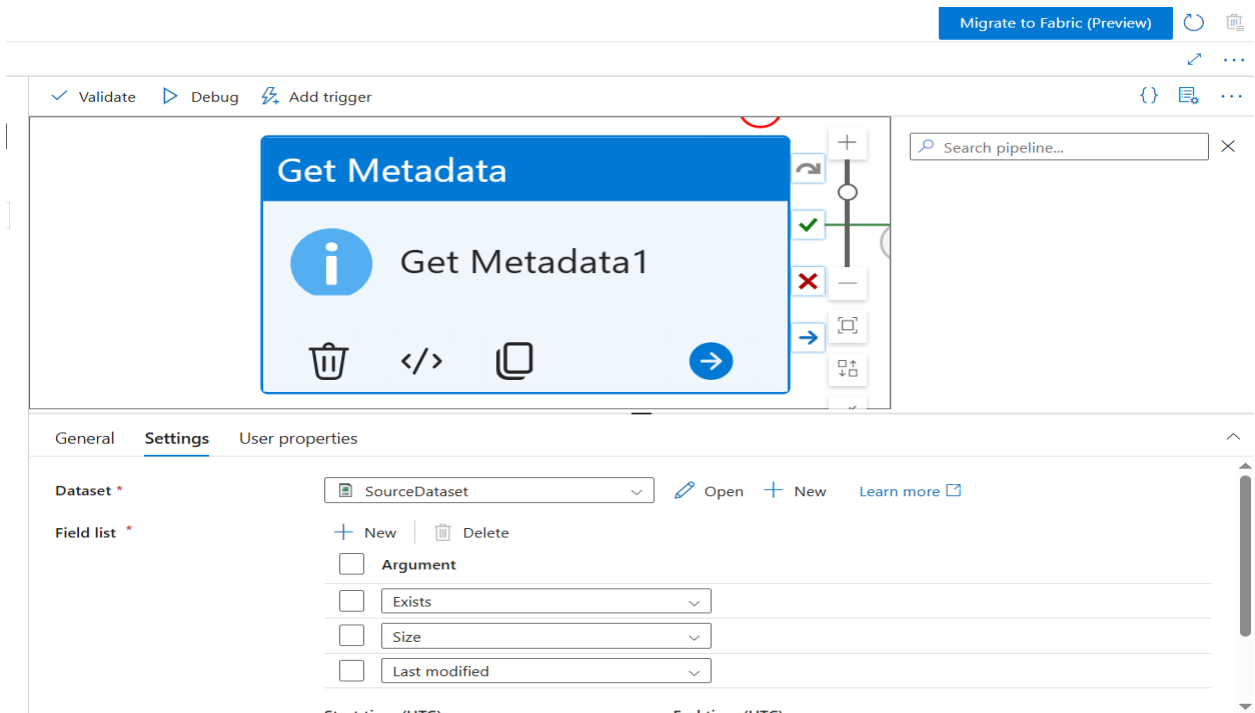
- **Use Get Metadata Activity**

The **Get Metadata** activity is used to retrieve metadata information about a file or dataset before performing further operations.

In this assignment, the activity was configured with the **SourceDataset** and the **Exists** property was selected to verify whether the source file is available before executing the copy operation.

Purpose of Get Metadata

- Check whether the source file exists.
- Validate the availability of data before processing.
- Improve pipeline reliability by avoiding execution errors.



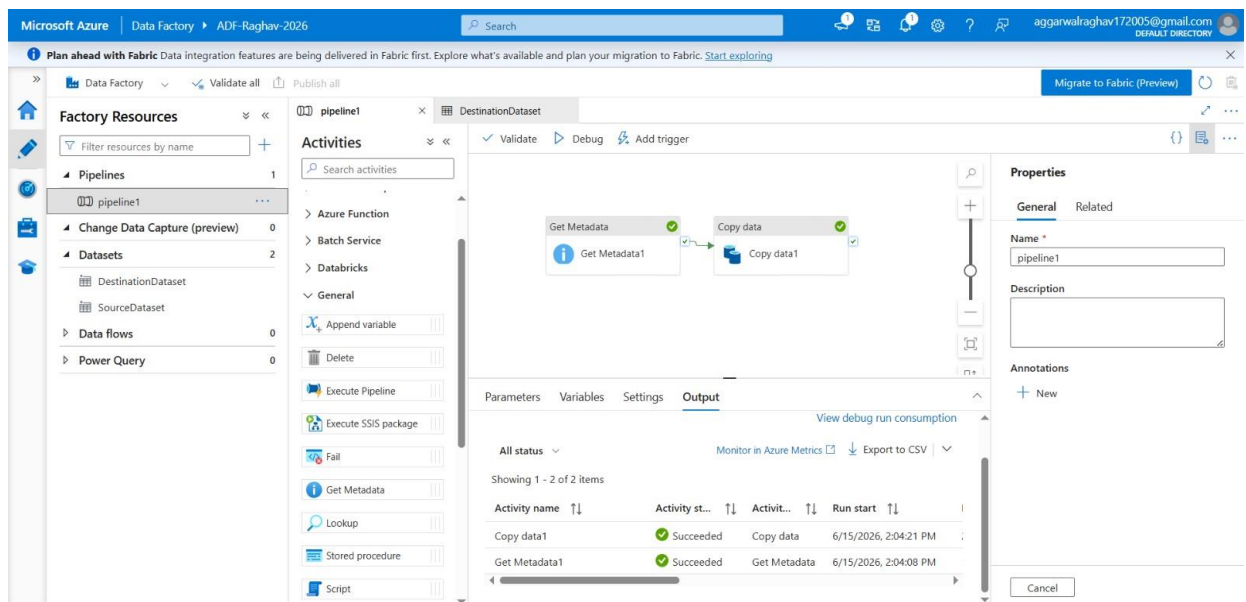
TASK 4: Pipeline Development

Objective

The objective of this task is to create an Azure Data Factory pipeline that copies data from a source dataset to a destination dataset using the **Copy Data** activity.

Pipeline Development

A pipeline named **CopyPipeline** was created in Azure Data Factory. The **Copy Data** activity was configured to read data from the **SourceDataset** (stored in the input container) and copy it to the **DestinationDataset** (stored in the output container).



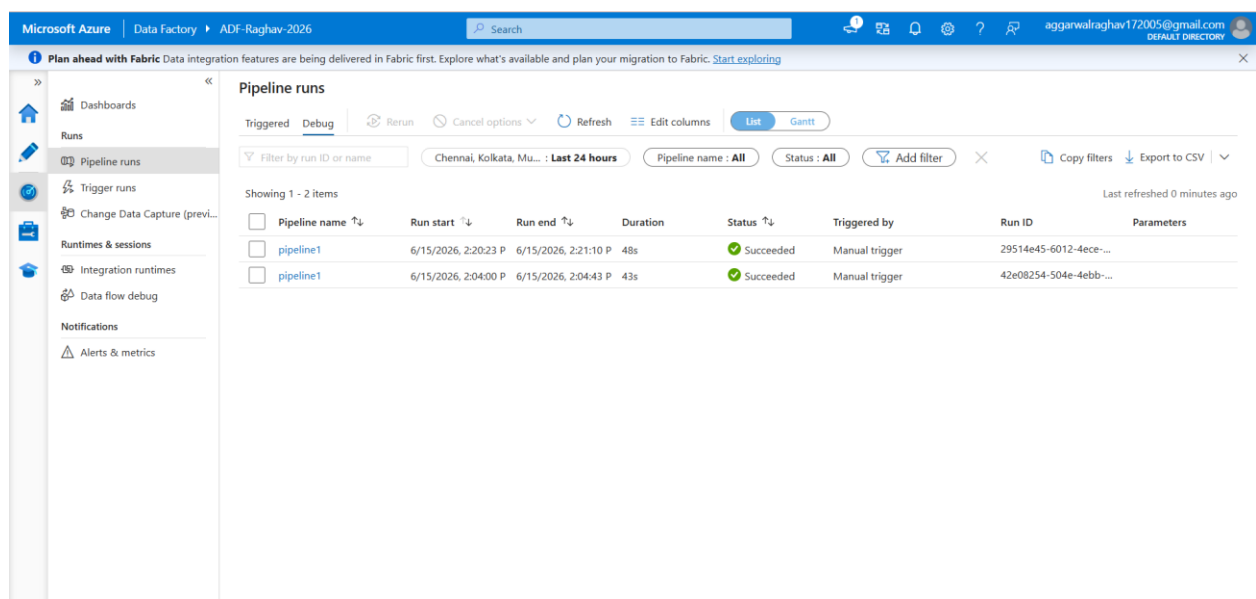
TASK 5: Pipeline Execution

Objective

The objective of this task is to execute the Azure Data Factory pipeline and verify that the data transfer process completes successfully.

Pipeline Execution

The pipeline was executed using the **Debug** option in Azure Data Factory. During execution, the **Get Metadata** activity first retrieved information about the source file, such as its existence, size, and last modified date. After successful validation, the **Copy Data** activity transferred the CSV file from the source container to the destination container. The execution status was monitored, and the pipeline completed successfully with a **Succeeded** status.



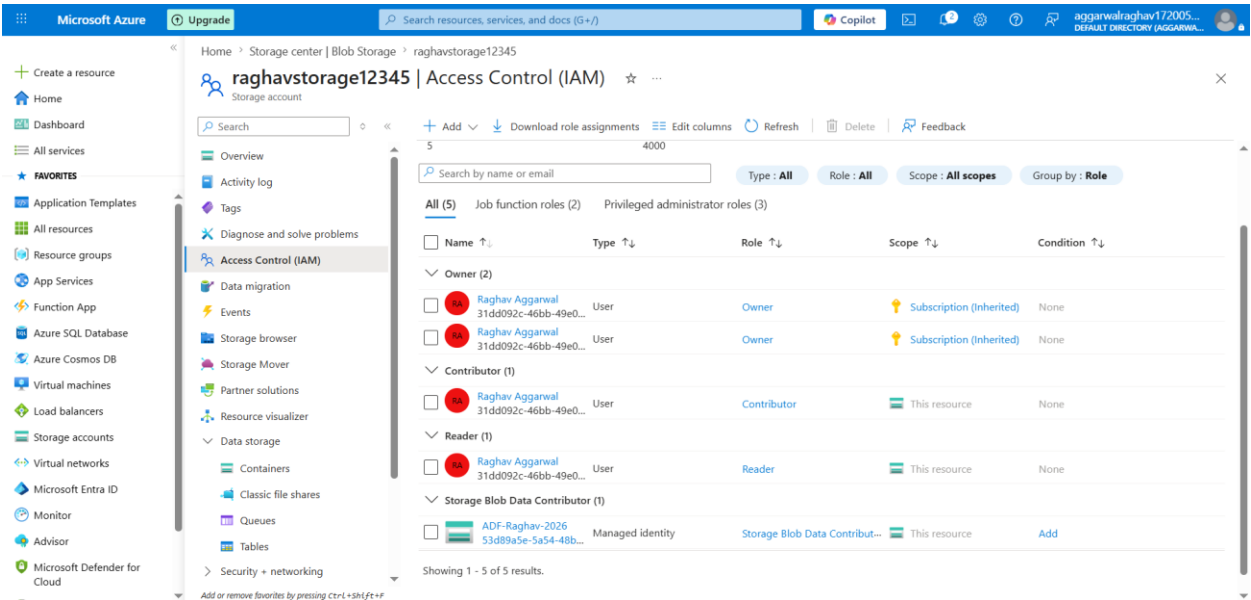
TASK 6: IAM Roles

Objective

The objective of this task is to manage access permissions by assigning appropriate IAM (Identity and Access Management) roles to users and Azure Data Factory.

Role Assignment

The **Reader** role was assigned to provide read-only access to the storage resources. Additionally, the **Storage Blob Data Contributor** role was assigned to Azure Data Factory to allow it to read and write data in Azure Blob Storage, enabling successful pipeline execution.



Conclusion

This assignment demonstrated the practical implementation of Azure Blob Storage and Azure Data Factory (ADF) for cloud-based data integration. A storage account and containers were created, linked services and datasets were configured, and the Get Metadata activity was used to retrieve file information. A pipeline was then developed using the Copy Data activity to transfer data from the source container to the destination container. Finally, appropriate IAM permissions were assigned to Azure Data Factory using the Storage Blob Data Contributor role. The successful execution of the pipeline confirmed that data was copied securely and efficiently within the Azure environment.